Invited Paper

Monte Carlo study of tunable negative-zero-positive index of refraction in nanosphere dispersed liquid crystals

W. Walasik[a], M. Jarema[a], G. Pawlik[a], R. Orlik[a], A.C. Mitus[a], F. Kajzar[b]

[a]Institute of Physics, Wroclaw University of Technology, Wybrzeze Wyspianskiego 27, Poland
[b] Laboratoire POMA CNRS, Universite d'Angers, Angers, France

ABSTRACT

Khoo et al.[1,2] have shown that nanosphere dispersed nematic liquid crystal (NDLC) constitutes a new type of metamaterial with index of refraction tunable from negative to positive values. Recently[3] we have combined this approach with Monte Carlo simulations of inhomogeneous molecular order in planar NLC cells. Lebwohl - Lasher effective hamiltonian with Rapini - Papoular term for anchoring forces was used. Electric field and amplitude of anchoring forces are control parameters which determine the profiles of order parameter. In this paper we study, using the same approach, local spatial distribution of refractive index in NDLC planar cell. We show that NDLC material consists of layers with negative–zero–positive index of refraction. The spatial organization of those layers strongly depends on incident light wavelength. The role of spatially modulated external electric field for tuning of refractive index of NDLC is briefly discussed.

Keywords: negative index of refraction, metamaterials, nematic liquid crystals, Monte Carlo

1. INTRODUCTION

The concept behind modern science of materials with negative refractive index was formulated by Veselago 40 years ago.[4] Since ten years metamaterials have been in center of interest because of their potential unusual optic and photonic applications. In this context, tunability of negative-zero-positive refractive index metamaterials is particularly desirable. Recently, this topic was addressed in a seminal paper of Khoo and co-workers.[1] It generalizes the approach of Wheeler at al.[5] by using nematic liquid crystal (NLC) as a medium in which the coated nanospheres are dispersed. The nanosphere–dispersed liquid crystal (NDLQ) model system displays negative real part of diffraction index at infrared frequencies. In paper[1] only uniform NLC order was studied. Successive applications of this approach were published in Refs.[6,7]

Inhomogeneous order in NLC host may be a direct cause of a non–uniform spatial distribution of negative refractive index, giving rise to tunable structure of inhomogeneous metamaterials. The simplest way to induce an inhomogeneous order in a planar NLC cell with anchoring forces is to apply either constant or modulated electric (magnetic) field.

Monte Carlo simulations enable a detailed "microscopic" study of inhomogeneous order in NLC systems. In particular, spatial distribution of refractive index can be calculated. The nature of orientational NLC order depends on amplitudes of two control parameters: anchoring force α and applied electric field E. The chess–board NLC order was studied in paper.[8] The phase diagram in $\alpha - E$ variables calculated in[9] displays a new line of transitions caused by anchoring forces. Further extensions of the method, in particular an application to twisted NLC displays, were presented in papers.[3,10,11]

Recently,[3] we have combined Monte Carlo simulations with the approach of Khoo[1] to calculate the phase diagram in $\alpha - E$ variables of averaged permittivity. We have shown that the amplitude of anchoring forces constitutes a new control parameter for tuning of negative–positive values of refraction index in NDLC.

The aim of this letter is to analyze the spatial inhomogeneity of refractive index in NDLC metamaterial for chosen values of incident light wavelength λ, anchoring force α and external electric field E.

Further author information: (Send correspondence to antoni.mitus@pwr.wroc.pl)

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2. THEORETICAL BACKGROUND

The effective refractive index of NDLC system was calculated using the approach introduced in papers.[1,5] Electromagnetic wave scattering from a single coated sphere was calculated using Mie theory;[12] the effective scattering from an ensemble of coated spheres dispersed in NLC matrix was analyzed in effective medium theory, using Maxwell Garnet mixing rule.[12–14] This approach relates an effective refractive index of NDLC metamaterial to the parameters of its constitutive components: coated nonmagnetic spheres, and host NLC system. The schematic shape of a single coated sphere is presented in Figure 1. Below, we present the basic formulas used for calculations.

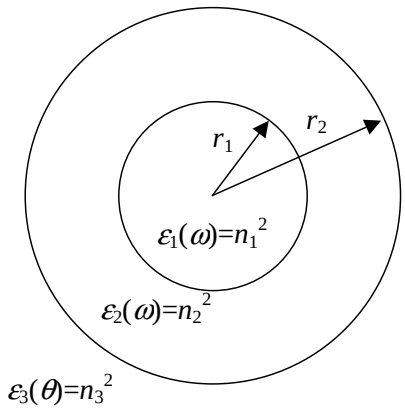

Figure 1. Parameters of the coated sphere (see text for details).

The core with radius r_1 is made from polaritonic material (such as $LiTaO_3$) with relative permittivity

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_T^2 - \omega_L^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right). \tag{1}$$

Here $\epsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively. Required relative permittivity is of the order of magnitude 10^2.[5] Bulk cavity resonance in the core is responsible for negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths.

The relative permittivity of the semiconductor shell with outer radius r_2 is described by Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient. The shell is responsible for negative effective permittivity, with ω_p close to frequency for which negative index will be detected. Its value is chosen to provide both negative effective permittivity and permeability in some interval of wavelengths.[5]

Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle θ between directions of incident light wave vector and director of NLC:[2]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are permittivities for parallel and perpendicular orientations of director with respect to incident wave vector $\vec{k}_0$ in vacuum. In pure NLC host (without spheres) $k_3 = \sqrt{\epsilon_3}\, k_0$.

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Electromagnetic wave Mie scattering was calculated in dipolar (electric and magnetic) approximations, with amplitudes of scattered fields proportional to coefficients a_1 and b_1, respectively. The detailed expressions can be found, e.g., in.[12] Those coefficients enter the expressions for effective permittivity and permeability of NDLC, calculated in effective medium theory:[1]

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{4}$$

$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}. \tag{5}$$

Those formulas are valid only for small number density $N = 3f/4\pi r_2^3$ of spheres (f is the filling factor) and in long-wavelength limit $2\pi/k_3 \gg r_2$. The effective index of refraction was calculated as

$$n_{eff} = \pm\sqrt{\epsilon_{eff}\ \mu_{eff}}, \tag{6}$$

where we have used the root with a positive imaginary part. Khoo et al. have shown[1] that for some values of the parameters of the model there is a range of frequencies ω and angles θ for which

$$n'_{eff}(\omega, \theta) \equiv \Re\big(n_{eff}(\omega, \theta)\big) < 0. \tag{7}$$

3. MONTE CARLO SIMULATION

The presentation in this Section is based on papers.[10,11] The equilibrium configurations of NLC molecules on a lattice were sampled using Metropolis Monte Carlo simulations[15,16] for Lebwohl–Lasher effective hamiltonian[17] with Rapini–Papoular term:[18]

$$H = -\xi \sum_{<\vec{r},\vec{r}\,'>} P_2(\cos\beta(\vec{r},\, \vec{r}\,')) - \sum_{\vec{r}} E^2(\vec{r}))\, P_2(\cos\beta(\vec{r})) + \sum_{\vec{r}_W} \alpha(\vec{r}_W)\, \sin^2\gamma(\vec{r}_W), \tag{8}$$

where P_2 denotes the second-order Legendre polynomial. Parameter ξ ($\xi = 25$ in 3D) is the coupling constant for NLC–NLC orientational interaction, $E(\vec{r})$) is external electric field, $\alpha(\vec{r}_W)$ – the amplitude of anchoring forces acting on molecules located at the walls. $\beta(\vec{r},\, \vec{r}\,')$ denotes the relative angle between two molecules located at points $\vec{r}$, $\vec{r}\,'$, $\beta(\vec{r})$ – the angle a molecule located at point $\vec{r}$ makes with electric field direction. $\gamma(\vec{r}_W)$ is the angle between long axis of molecule at the wall and a fixed rubbing direction. The rubbing direction is parallel to x axis (Fig. 2).

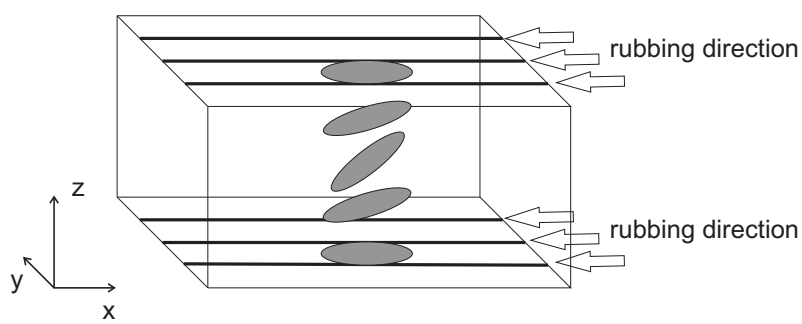

Figure 2. Geometry of the model.

The simulations of NLC were performed on a cubic $50 \times 50 \times 20$ lattice. An orientation of a long axis of a NLC molecule in an external coordinate system is described by a pair of angles ϕ, θ. Angle ϕ (twist) describes an azimuth and angle θ (tilt) is a polar angle calculated from z–axis, along which the electromagnetic wave propagates.

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4. INOMOGENEOUS METAMATERIAL IN PLANAR NDLC CELL

Spatial modulation of applied external electric field and spatial modulation of the strength of anchoring forces, the later deterministic or random in nature, has a strong impact on resulting inhomogeneous order in planar NLC cell.[3,8–11] It determines the spatial distribution of refractive index which, for some values of parameters, can assume negative values. Since negative refraction index requires a fine tuning of the parameters of NDLC,[1] the specific details of local order inhomogeneity can play a decisive role for its coming into existence and, in particular, for the character of its spatial organization.

In present study we address the simplest case when (i), external electric field is constant: $\vec{E}(\vec{r}) = E\,\hat{z}$ and (ii), anchoring forces are spatially homogeneous: $\alpha(\vec{r}_W) = \alpha = const$. Even in this simple case a compound metamaterial is shown to exist.

Fig. 3 shows the functional dependence of real part of the (local) refractive index n'_{eff} on NLC particle orientation (polar angle θ), calculated for parameters used by Khoo et al.:[1] $r_1 = 0.13\ \mu\mathrm{m}, \mathrm{r}_2 = 0.143\ \mu\mathrm{m}, \mathrm{f} = 0.12, \epsilon(\infty) = 17, \omega_\mathrm{T}/2\pi = 240$ THz, $\omega_L/2\pi = 570$ THz, $\gamma_1/2\pi = 2.5$ THz, $\omega_\mathrm{p}/2\pi = 134$ THz and $\gamma_2 = \omega_p/60$. Two close values of wavelengths are investigated: $\lambda = 2.820\ \mu$m and $2.850\ \mu$m, both from the interval of wavelengths studied in.[1] In spite of close values of λ the plots $n'_{eff}(\theta)$ display very different behavior for small angles $\theta < 25°$ which correspond to approximate parallelism of directions of local director and of external electric field. Namely, for linearly polarized light with $\lambda = 2.820\ \mu$m incident as an extraordinary wave on NLC host, NDLC forms a metamaterial with $n'_{eff} < 0$. On the contrary, light with $\lambda = 2.850\ \mu$m propagates in a system with positive refractive index. For large values of angle $\theta > 60°$ both refractive indices are positive.

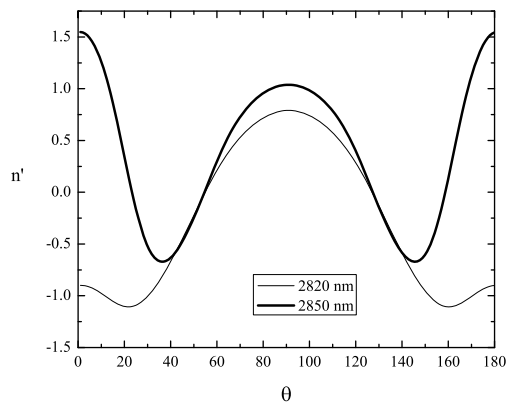

Figure 3. Real part of the local refractive index vs. NLC particle orientation (θ angle).

We have simulated the system in the strong anchoring regime[3,11] with $\alpha = 50$ and for electric field $E = 1.2$ exceeding the Freedeericksz threshold.[10] Fig. 4 shows the cross–section in $x-z$ plane of 3D configuration of NLC molecules. Strong anchoring forces keep the molecules adjacent to walls parallel to them while in the middle of the cell NLC molecules are nearly parallel to electric field direction.

Two–dimensional maps of spatial distribution of $n'_{eff}(x, y, z)$ averaged over y axis are shown in Fig. 5. We have neglected, for simplicity, angle ϕ and have taken into account only polar angle θ. Regions marked with striped pattern have negative values of refractive index. The distributions do not depend on x coordinate because of absence of electric field modulation along x direction. After averaging over x and y directions we get profiles $n'_{eff}(z)$ presented in Fig. 6.

The NDLC displays the layered structure with layers of normal material with $n'_{eff}(z) > 0$ and of metamaterial with $n'_{eff}(z) < 0$. The spatial organization of the layers strongly depends on incident light wavelength. For $\lambda = 2.820\ \mu$m we find three layers with central one displaying metamaterial properties. In the case of $\lambda = 2.850\ \mu$m we observe five layers; two of them correspond to a metamaterial.

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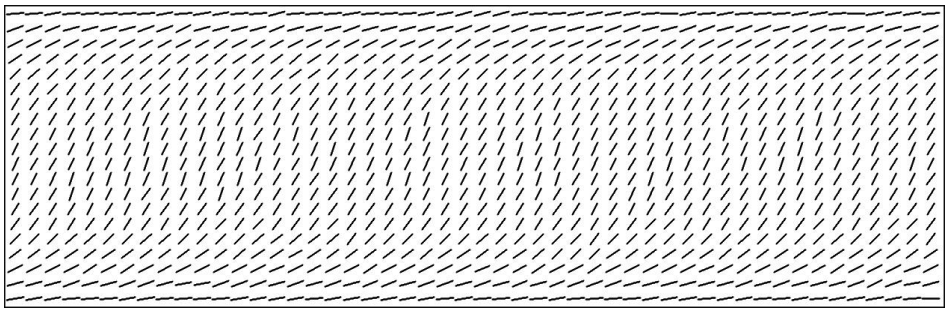

Figure 4. Cross-section in $x - z$ plane of 3D configuration of NLC particles for $E = 1.2$ and $\alpha = 50$.

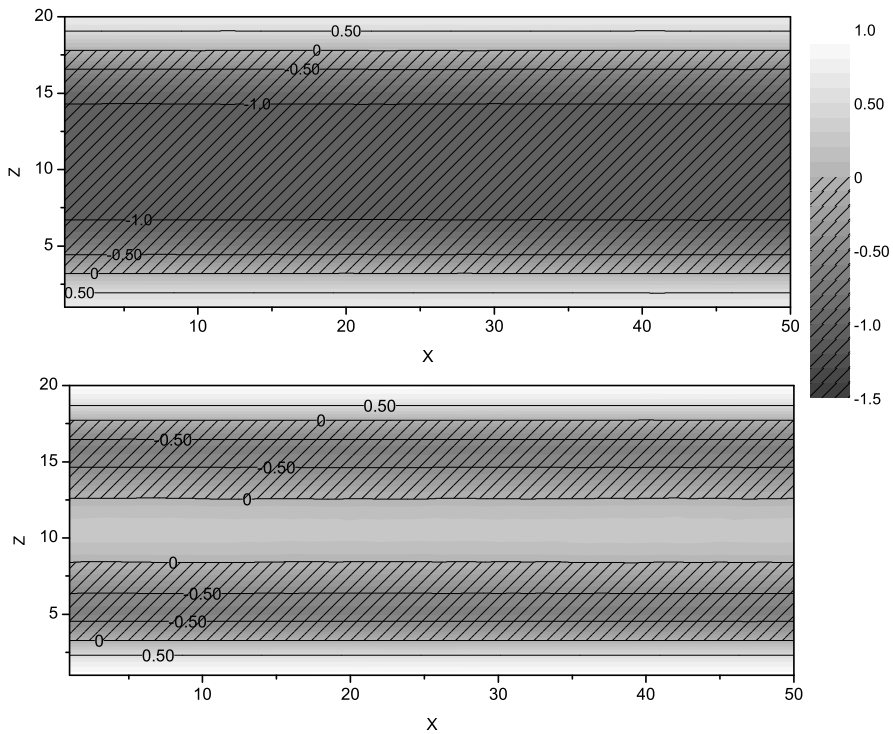

Figure 5. 2D maps of distribution of $n'_{eff}(x, y, z)$ averaged over y axis. Regions marked with striped pattern have negative values of refractive index. $\lambda = 2.820$ μm (top), $\lambda = 2.850$ μm (bottom).

5. CONCLUSIONS

We have applied lattice Monte Carlo simulations to the nanosphere dispersed liquid crystal model proposed recently by Khoo et al.[1] The studied system was a planar cell in a constant external electric field and in the strong anchoring forces regime. The inhomogeneous NLC order gave rise to a spatially inhomogeneous distribution of refractive index. Optical properties of the system were studied for two wavelengths $\lambda = 2.820$ μm and 2.850 μm of linearly polarized light incident on the system as extraordinary wave.

We have shown that the NDLC consists of layers of normal material which alternate with layers of metamaterial. Even a small variation of incident light wavelength leads to very different inhomogeneous spatial organization of metamaterial. This effect comes from sophisticated relation between refractive index and incident light wavelength. Demonstrated effect of electric field tuning of spatial organization of NDLC can be of interest for photonic applications. Further studies oriented onto a systematical analysis of influence of electric

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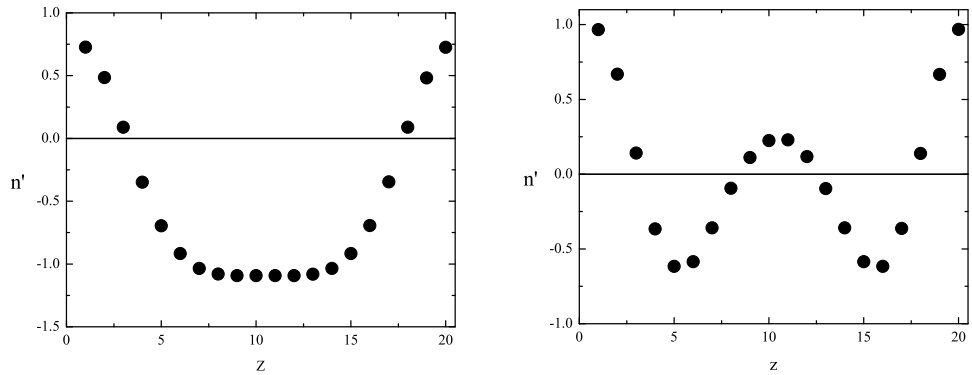

Figure 6. Profiles $n'_{eff}(z)$ calculated for $E = 1.2$, $\alpha = 50$. $\lambda = 2.820$ μm (left); $\lambda = 2.850$ μm (right).

field amplitude and of strength of anchoring force on spatial organization of normal material and metamaterial is under progress now. We point out, in this context, that spatial modulation of external electric field may lead to a very complex organization of spatial structures of metamaterial. This topic is at progress now.

We have analyzed the 2D maps of spatial distribution of refractive index only. However, the transmission and absorption coefficients for such a system can also be calculated. For strictly layered systems those topics were reported in papers.[19–21]

Similar studies for an interval of values of anchoring force and of applied electric field are necessary to calculate the phase diagram of negative-zero-positive index of refraction in variables $\alpha - E$, allowing for a potentially new method of tuning of optical properties of NDLC. First results for permittivity were presented in paper.[3] This work is at progress now.

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REFERENCES

1. I.C. Khoo, D.H. Werner, X. Liang, and A. Diaz, Opt. Lett. **31**, 2592 (2006).
2. I.C. Khoo, *Liquid crystals*, Wiley, 2007.
3. G. Pawlik, A.C. Mitus, and F. Kajzar, Proc. SPIE **6654**, 66540G (2007).
4. V.G. Veselago, Sov. Phys. Usp. **10**, 509 (1968).
5. M.S. Wheeler, J.S. Aitchison, and M. Mojahedi, Phys. Rev. B **73**, 045105 (2006).
6. D.H. Werner, D.H. Kwon, I.C. Khoo, A.V. Kildishev, and V.M. Shalaev, Opt. Expr. **15**, 3342 (2007)
7. I.C. Khoo, A. Diaz, S. Kubo, J. Liou, M. Stinger, T. Mallouk, and J.H. Park, Mol. Cryst. Liq. Cryst. **485**, 186 (2008).
8. G. Pawlik, A.C. Mitus, and A. Miniewicz, Opt. Commun. **182** (2000) 249-254.
9. K. Komorowska, G. Pawlik, A.C. Mitus, and A. Miniewicz, J. Appl. Phys. **90** (2001) 1836-1842.
10. G. Pawlik, A.C. Mitus, and F. Kajzar, Proc. SPIE **5990**, 59900F (2005).
11. G. Pawlik, A.C. Mitus, and F. Kajzar, Proc. SPIE **6332**, 63320B (2006).
12. C.F. Bohren and D.R. Huffman, *Absorption and Scattering of Light by Small Particles*, Wiley, 1998.
13. J.C. Maxwell Garnett, Philos. Trans. R. Soc. Ser. A **203**, 385 (1904).

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14. C.A. Guérin, P. Mallet, and A. Sentenac, J. Opt. Soc. Am. A **23**, 349 (2005).
15. P. Allen and D. Tildesley, *Computer Simulation of Liquids*, Clarendon Press, Oxford, 1987.
16. D. Landau and K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, Cambridge University Press, 2000.
17. P.A. Lebwohl and G. Lasher, Phys. Rev. A **6**, 426 (1972).
18. A. Rapini and M. Papoular, J. Phys. (France) **30**, 54 (1969).
19. T. Pan, C. Tang, L. Gao, and Z. Li, Phys. Lett. A **337**, 473 (2005).
20. Y. Dong and X. Zhang, Phys. Lett. A **359**, 542 (2006).
21. R. Aylo, P.P. Banerjee, and G. Nehmetallah, Proc. SPIE **7029**, 702917 (2008).

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